

Project report on

Quality Care and its Cost in Health Informatics: Examining Socioeconomic Status (SES)

and Health Outcomes

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1. Introduction

Healthcare disparities rooted in socioeconomic status (SES) represent one of the most persistent challenges in modern health systems (Barakat & Konstantinidis, 2023). While medical advancements have improved population health overall, the benefits have not been distributed equally across social classes. This paper examines the critical intersection of health informatics, quality of care, and cost management through the lens of socioeconomic disparities. The relationship between SES and health outcomes is well-established: individuals with lower socioeconomic status consistently experience poorer physical and mental health outcomes, reduced access to preventive services, and higher mortality rates compared to their higher-SES counterparts (Hamasaki, 2025). What makes this issue particularly urgent in 2024-2025 is the documented reversal of a 50-year trend wealthier Americans now receive significantly more healthcare than middle- and lower-income populations, despite often having fewer health needs (Larnyo et al., 2025).

Health informatics the intersection of information technology, healthcare, and data management offers transformative potential to address these disparities (Javaid et al., 2024). By leveraging electronic health records (EHRs), telehealth platforms, predictive analytics, and health information exchanges (HIEs), informatics tools can improve care coordination, reduce costs, and enhance quality across all socioeconomic strata (Kruse et al., 2025). However, the effectiveness of these technologies depends critically on addressing the structural barriers that prevent equitable access and utilization (Xu et al., 2025). This research focuses on three core questions: (1) How does SES influence access to and quality of health informatics interventions? (2) What are the cost implications of digital health disparities? and (3) How can informatics solutions be designed to promote rather than perpetuate health equity?

2. Topic's Impact on the Population/Public Health Field

The integration of health informatics into SES-focused care delivery has profound implications for population health:

2.1 Technology as a Disparity Mitigation Tool

Recent evidence demonstrates that technology access and healthcare technology use significantly reduce healthcare disparities. A 2025 study analyzing 815 African Americans found that technology access had a significant negative effect on healthcare disparity ($\beta = -0.086$, $p = 0.034$), while healthcare technology use showed an even stronger negative effect ($\beta = -0.180$, $p < 0.001$) (Larnyo et al., 2025). Notably, SES moderated the relationship between technology access and disparities, indicating that enhanced technology access particularly benefits lower-SES populations (Larnyo et al., 2025).

2.2 Economic Efficiency and Cost Savings

Digital health interventions (DHIs) demonstrate substantial cost-effectiveness across chronic disease management. A systematic review of 22 economic evaluations found that 82% of DHIs for children and adolescents were cost-effective or cost-saving compared to standard care, with incremental cost-effectiveness ratios ranging from \$24 to \$23,602 per DALY averted (Degli Esposti et al., 2023). For adult populations with chronic diseases, SMS-based interventions for diabetes management achieved cost savings of \$437 per participant while improving HbA1c levels by 0.4% (Yang et al., 2023).

2.3 Emergency Department Quality Improvements

Health information technology specifically enhances quality of care in emergency departments settings where SES disparities are often most acute. A 2025 systematic review found

that HIT implementation in emergency departments improved care coordination, reduced medication errors, and enhanced clinical decision-making, particularly benefiting populations with complex social needs who frequently utilize emergency services (Kruse et al., 2025).

2.4 Population Health Management

Data analytics enables healthcare providers to identify at-risk patient groups and analyze community health trends, including monitoring social determinants of health (SDOH) such as economic stability, education access, and community environment (Javaid et al., 2024). By integrating SDOH data into EHRs, health systems can create targeted interventions for vulnerable populations, shifting from reactive to proactive care models (Javaid et al., 2024).

3. Challenges Associated with the Chosen Research Area

Despite the potential of health informatics to reduce disparities, significant challenges persist:

3.1 The Digital Divide and Technology Access

The digital divide remains the primary barrier to equitable health informatics implementation. Only 15% of healthcare AI tools include community engagement in development, and 85% of AI health equity studies track outcomes for less than 12 months, limiting understanding of long-term impacts (Kraus et al., 2025). Algorithmic bias leads to 17% lower diagnostic accuracy for minority patients, while the digital divide excludes 29% of rural adults from AI-enhanced healthcare tools (Kraus et al., 2025). Larnyo et al. (2025) found that SES significantly affects technology access ($\beta = 0.424$, $p < 0.001$), creating a cascade where lower-SES individuals have reduced access to the very tools that could mitigate their health disparities.

3.2 Interoperability and Data Exchange Barriers

Hospitals serving disadvantaged communities face systematic barriers to health information exchange. In 2023, 62% of US hospitals experienced at least one major barrier to interoperable exchange, with lower-resourced hospitals experiencing significantly more major barriers than their higher-resourced counterparts (2.07 vs. 1.48 for large hospitals) (Holmgren et al., 2025). Common barriers include different vendor platforms limiting data exchange, high costs of exchanging data with outside organizations, patient matching difficulties between systems, and contractual constraints between vendors and healthcare providers (Holmgren et al., 2025). Rural and independent hospitals often serving lower-SES populations experience even greater challenges, with an average of 2.27 major barriers compared to 1.63 for system-affiliated hospitals (Holmgren et al., 2025). This disparity perpetuates care fragmentation, as patients receiving treatment at these facilities are less likely to have their health information seamlessly shared across providers. The resulting information gaps disproportionately affect lower-SES patients who already face greater difficulty navigating complex healthcare systems, ultimately reinforcing the very disparities that health informatics seeks to address.

3.3 Socioeconomic Status and Health Technology Adoption

Xu et al. (2025) found that technological challenges disproportionately affect people with lower SES, creating a "double burden" where those with greatest health needs have least access to informatics solutions. Hamasaki (2025) further documented that SES influences health behaviors in older adults, with lower-SES populations showing reduced engagement with digital health tools due to limited health literacy and technology comfort.

3.4 Algorithmic Bias and AI Equity

AI systems trained on non-representative data systematically underperform for underrepresented groups. A widely-used algorithm for predicting healthcare needs underestimated

Black patients' illness severity by approximately 48% because it used healthcare costs as a proxy for health needs, failing to account for systemic inequities in care access (Obermeyer et al., 2019, as cited in NAACP, 2025). In dermatology AI, training datasets historically include less than 15% representation of darker skin tones despite comprising 40% of the US population, creating performance gaps of 20-50% for certain conditions (NAACP, 2025).

3.5 Infrastructure and Policy Gaps

In developing countries, health informatics faces additional challenges including limited technical infrastructure, workforce shortages, and lack of established policies (Soyege et al., 2025). The Federal Health IT Strategic Plan 2024-2030 identifies that expanding broadband access and ensuring affordable connected devices remain critical unresolved issues for underserved populations (Office of the National Coordinator for Health IT, 2024).

4. Related Work by Others Conducting Research in This Area

4.1 SES-Health Relationship Frameworks

Barakat and Konstantinidis (2023) provided a comprehensive systematic review of socioeconomic status change and health outcomes, establishing that social mobility whether upward or downward significantly impacts health trajectories across the life course. Their analysis revealed that individuals experiencing downward social mobility show worse health outcomes than those with stable SES, while upward mobility is associated with improved health, though often with time lags. This dynamic framework is particularly relevant for health informatics research because it suggests that interventions must account for SES volatility rather than treating socioeconomic status as a static variable. Their work directly informs understanding of how informatics interventions might interrupt negative health cycles by providing timely support during periods of socioeconomic transition. Furthermore, they identified mechanisms including stress

pathways, behavioral changes, and access barriers that mediate the SES-health relationship mechanisms that digital health tools are uniquely positioned to address through continuous monitoring and adaptive support systems.

4.2 Technology Access and Disparities

The intersection of technology access, socioeconomic status, and healthcare disparities has emerged as a critical research domain. Larnyo et al. (2025) employed Partial Least Squares-Structural Equation Modeling to analyze data from 815 African American adults, demonstrating that SES significantly moderates the relationship between technology access and healthcare disparities ($\beta = 0.424$, $p < 0.001$). Their findings indicate that while technology access reduces disparities overall, the magnitude of benefit varies substantially by socioeconomic level lower-SES individuals gain greater relative reduction in disparities from equivalent technology access, suggesting that equitable deployment of digital health tools could have disproportionate positive impact on disadvantaged populations. This methodological framework provides a template for future informatics equity research, particularly in demonstrating how structural equation modeling can disentangle direct effects from moderating influences in complex health technology implementations. Livingood et al. (2022) conducted comparative studies of neighborhood clinics across different SES levels, finding significant gradients in health literacy and internet access that directly impact digital health intervention effectiveness. Their research documented that clinics serving lower-SES neighborhoods had 34% lower rates of patient portal enrollment and 28% lower rates of telehealth utilization, even when technology infrastructure was equivalent. These findings highlight that access to technology infrastructure alone is insufficient; effective informatics interventions must address health literacy barriers, digital skills acquisition, and trust in digital systems factors that vary systematically by socioeconomic status.

4.3 Health Informatics Effectiveness in Clinical Settings

Kruse et al. (2025) systematically reviewed health information technology effects in emergency departments, synthesizing evidence from 47 studies to establish the clinical context for SES-focused informatics applications. Their analysis found that HIT implementation improved care coordination scores by an average of 23%, reduced medication errors by 31%, and decreased average length of stay by 0.8 days—benefits that are particularly significant for lower-SES populations who rely more heavily on emergency services for primary care. However, they noted that these benefits were attenuated in safety-net hospitals with higher proportions of uninsured patients, suggesting that implementation context mediates effectiveness. This work establishes that while informatics can improve quality, realizing these benefits equitably requires targeted attention to resource-constrained settings. Javaid et al. (2024) provided an extensive analysis of health informatics features, contributions, applications, and limitations, noting that cultural adaptation remains a critical gap in current systems. Their review identified that 78% of commercially available health informatics platforms lack culturally tailored interfaces, and only 12% incorporate multilingual support beyond English and Spanish. These limitations disproportionately affect lower-SES populations, who are more likely to be linguistically isolated or from non-dominant cultural backgrounds. Their work underscores that technical functionality must be matched with cultural appropriateness to achieve equitable outcomes.

4.4 Economic Evaluation of Digital Health Interventions

Multiple systematic reviews have established the economic value of digital health interventions, though with important caveats regarding generalizability across SES levels. Degli Esposti et al. (2023) reviewed 22 economic evaluations of digital health interventions for children and adolescents, finding that 82% achieved cost-effectiveness thresholds (defined as <\$50,000 per

QALY gained). However, they noted that studies rarely reported SES-stratified results, and those that did showed smaller effect sizes in lower-SES subgroups due to lower engagement rates. Key cost-effectiveness drivers identified include population coverage (penetration rates), implementation costs (including staff training and technical support), and intervention effect size all of which vary significantly by socioeconomic context. Yang et al. (2023) conducted a systematic review specifically examining cost-effectiveness of digital tools for chronic disease management among adults, finding that 70% of apps achieved cost-effectiveness thresholds. Notably, SMS-based interventions showed the strongest cost-effectiveness profiles (\$437 cost savings per participant for diabetes management) and the smallest SES-based disparities in outcomes, suggesting that lower-complexity interventions may be more equitable. Their work indicates that the cost-quality trade-off in health informatics must be evaluated not only at the population level but across SES strata to ensure that efficiency gains do not come at the expense of equity.

4.5 Global Health Informatics and Resource-Limited Contexts

Soyege et al. (2025) examined health informatics in developing countries, identifying challenges including limited technical infrastructure, workforce gaps, and lack of established policies factors that are equally relevant to underserved populations in high-income countries. Their comparative analysis revealed that resource-limited settings often leapfrog to mobile-based solutions, bypassing desktop-dependent systems that dominate in well-resourced contexts. This "leapfrog" pattern suggests that lower-SES populations in the US might benefit from mobile-first informatics designs rather than adaptations of enterprise systems designed for well-resourced settings. Their work on infrastructure constraints, training requirements, and policy needs provides

a framework for understanding barriers in US "digital deserts" rural and urban areas with limited broadband access that mirror developing country contexts.

4.6 Artificial Intelligence and Health Equity

Recent work has established frameworks for operationalizing AI for health equity, moving beyond technical accuracy to fairness metrics. The NAACP (2025) published comprehensive guidelines for designing AI for health equity, including bias audit toolkits (Aequitas, Fairlearn) and governance structures necessary for equitable deployment. Their framework emphasizes that algorithmic fairness requires explicit attention to subgroup performance, with recommendations that AI tools demonstrate <5% performance gap across racial and SES groups before deployment. This represents a significant shift from traditional accuracy-focused validation and provides concrete standards for equity-centered informatics development. Kraus et al. (2025) extended this work by examining AI as a catalyst for health equity in primary care, finding that only 15% of current healthcare AI tools include community engagement in development, and 85% of AI health equity studies track outcomes for less than 12 months. Their analysis documented that algorithmic bias results in 17% lower diagnostic accuracy for minority patients and that the digital divide excludes 29% of rural adults from AI-enhanced healthcare tools. These findings establish that without deliberate equity-focused design, AI risks automating and amplifying existing disparities rather than reducing them.

4.7 Synthesis of Related Work

Collectively, this body of research establishes several critical foundations for the current study. First, the SES-health relationship is dynamic and mediated by multiple mechanisms that informatics can potentially address, but only if designed with socioeconomic context as a core consideration rather than an afterthought (Barakat & Konstantinidis, 2023; Larnyo et al., 2025).

Second, technology access effects are moderated by SES, suggesting that equitable deployment strategies must prioritize lower-SES populations to maximize population health impact (Larnyo et al., 2025; Livingood et al., 2022). Third, while health informatics demonstrates clear effectiveness in improving quality and reducing costs, these benefits are not automatically distributed equitably and may be attenuated in resource-constrained settings without specific adaptation (Kruse et al., 2025; Javaid et al., 2024). Fourth, economic evaluations must incorporate SES-stratified analyses to ensure that cost-effectiveness at the population level does not mask widening disparities (Degli Esposti et al., 2023; Yang et al., 2023). Finally, emerging AI technologies require explicit equity governance frameworks to prevent algorithmic bias from perpetuating or amplifying SES-based disparities (NAACP, 2025; Kraus et al., 2025). The current research builds upon these foundations by specifically examining how the cost-quality trade-off in health informatics varies across socioeconomic strata, addressing the gap in SES-stratified economic evaluation identified by prior work. By integrating insights from technology access research, clinical effectiveness studies, and AI equity frameworks, this study aims to inform the design of informatics interventions that improve quality and manage costs equitably across all SES levels.

5. Opportunities for New Areas of Research

5.1 Synthetic Data for Algorithmic Equity

Synthetic data generation represents a paradigm shift for addressing representation gaps in AI training datasets. Artificially generated patient records can balance underrepresented groups while preserving privacy, potentially reducing diagnostic accuracy disparities without exposing real patient data (NAACP, 2025). Future research should validate synthetic data effectiveness across diverse clinical conditions and care settings.

5.2 Federated Learning for Privacy-Preserving Equity Research

Federated learning training AI models across decentralized data sources without centralizing sensitive patient information offers opportunity to include underrepresented populations in algorithmic development while addressing privacy concerns (NAACP, 2025). This approach could enable multi-institutional collaboration on SES-specific model development without data sharing barriers.

5.3 SDOH-Integrated Predictive Analytics

The integration of social determinants of health into machine learning-driven decision support systems remains underdeveloped. The PRADS (Predictive Risk Analytics for Diabetes Support) protocol represents an emerging model for incorporating SDOH data into clinical workflows, but broader application across chronic conditions and acute care settings requires further investigation (Zhang et al., 2024).

5.4 Tele-AI for Rural and Underserved Access

AI-enhanced telehealth platforms show promise for extending specialist care to remote areas. Research opportunities include developing AI diagnostic support for primary care clinicians serving rural populations, intelligent remote monitoring systems that account for technology access limitations, and virtual health assistants designed for low health literacy populations (Kraus et al., 2025).

5.5 Value-Based Payment Model Integration

Accountable Care Organizations (ACOs) show higher adoption of telehealth and HIE functionalities even in disadvantaged areas, suggesting that payment reform can drive technology equity (Holmgren & Milstein, 2024). Research should examine how alternative payment models can incentivize informatics investments in underserved communities and whether these investments translate to measurable disparity reduction.

5.6 Long-Term Equity Impact Assessment

With 85% of current AI health equity studies tracking outcomes for less than 12 months (Kraus et al., 2025), longitudinal research examining multi-year impacts of informatics interventions on SES-based health disparities represents a critical gap. Such studies should assess whether digital health tools produce sustained behavior change or if engagement disparities widen over time.

6. Overall Assessment and Conclusion

Health informatics stands at a critical juncture regarding socioeconomic health disparities. The evidence demonstrates both significant potential and substantial risks: while technology access and use can reduce healthcare disparities (Larnyo et al., 2025), current implementation patterns often exacerbate existing inequities through algorithmic bias, interoperability barriers, and the digital divide (Kraus et al., 2025; Holmgren et al., 2025). This tension between promise and peril defines the current research landscape and demands careful navigation to ensure that technological advancement serves equity goals rather than undermining them.

The research landscape reveals several key insights that should guide future development and policy. First, SES operates as both a direct determinant of health outcomes and a moderator of technology effectiveness. Lower-SES populations face compounding disadvantages—greater health needs, reduced technology access, and systems not designed for their contexts—that informatics solutions must explicitly address rather than assume universal benefit (Larnyo et al., 2025; Xu et al., 2025). This compounding effect means that interventions designed for average populations often fail to reach those with the greatest need, widening rather than narrowing disparities.

Second, cost-effectiveness is achievable but context-dependent. While 82% of digital health interventions demonstrate economic value (Degli Esposti et al., 2023), success depends on population coverage, implementation costs, and effect size—factors heavily influenced by SES-related engagement patterns (Degli Esposti et al., 2023). Interventions must be designed for the specific needs, capabilities, and preferences of lower-SES populations to achieve projected returns. Without such targeted design, the economic benefits of health informatics may accrue primarily to higher-SES populations who are already healthier, exacerbating the trend of wealthier Americans receiving more care despite fewer needs.

Third, structural barriers persist despite policy efforts. The Federal Health IT Strategic Plan 2024-2030 emphasizes equity (Office of the National Coordinator for Health IT, 2024), yet interoperability barriers remain more prevalent at lower-resourced hospitals (Holmgren et al., 2025), and algorithmic bias continues to affect diagnostic accuracy for minority patients (Kraus et al., 2025). This gap between policy aspiration and implementation reality indicates that technical solutions alone cannot overcome systemic underinvestment in health IT infrastructure for underserved communities. Addressing these barriers requires sustained resource allocation and accountability mechanisms that prioritize equity metrics alongside traditional quality measures.

Fourth, emerging technologies offer both promise and peril. AI and machine learning can enhance risk stratification and enable proactive interventions (Zhang et al., 2024), but without deliberate equity-focused development, these tools risk automating existing biases at scale (NAACP, 2025). The use of synthetic data, federated learning, and bias audit frameworks represents necessary but insufficient progress toward equitable AI. These technical approaches must be paired with community engagement, diverse development teams, and ongoing equity

monitoring to prevent the amplification of historical disparities through algorithmic decision-making.

Addressing SES-based health disparities through informatics requires moving beyond "equal access" frameworks to "equitable design" approaches (NAACP, 2025). This paradigm shift means developing technologies specifically for lower-SES contexts rather than adapting systems designed for advantaged populations, integrating social determinants of health data into clinical workflows to enable proactive rather than reactive care (Zhang et al., 2024), implementing value-based payment models that incentivize equity-focused technology adoption (Holmgren & Milstein, 2024), and establishing governance structures that prioritize fairness metrics alongside clinical accuracy (Kraus et al., 2025). Future research must prioritize longitudinal equity impact assessment, moving beyond the current pattern where 85% of AI health equity studies track outcomes for less than 12 months (Kraus et al., 2025). Additionally, community-engaged development processes and validation of emerging technical approaches across diverse socioeconomic contexts are essential to ensure that innovations serve broad population health goals.

The cost of inaction is high: as wealthier populations increasingly utilize advanced health technologies while lower-SES communities face persistent barriers, informatics risks becoming a force for health stratification rather than equalization (Larnyo et al., 2025). This reversal of the historical trend toward more equitable care distribution threatens to embed socioeconomic disparities more deeply into the healthcare system through technological mechanisms that are difficult to perceive and challenge. Proactive, equity-centered research and policy intervention can ensure that health informatics fulfills its potential to improve quality and manage costs for all populations regardless of socioeconomic status. The window for shaping this trajectory is narrow;

as AI and digital health systems become more deeply embedded in care delivery, the opportunity to design for equity diminishes while the cost of retrofitting for fairness increases exponentially.

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